locally $P \cong M \times G$ $\qquad \pi(u) = m$

$\rightarrow T_u P \cong T_m M \oplus \mathfrak{g}$

$\mathfrak{g} \cong V_u \qquad T_m M \cong H_u$

---

local trivialisation $\quad \phi : \pi^{-1}(U) \subseteq P \rightarrow U \times G \qquad \phi(p) = (u, g)$

$$d\phi_p : T_p P \rightarrow T_u U \oplus T_g G$$

$X \in T_p P \qquad f : U \times G \rightarrow \mathbb{R}$

$d\phi_p(X) f = X(f \circ \phi)$

$$\boxed{A^*_p \in T_p P \qquad A^*_p h = \frac{d}{dt}\Big|_{t=0} \left(h(p \triangleleft e^{tA})\right)}$$ definition

Def: $T_p P = \{$set of derivations at $p\}$

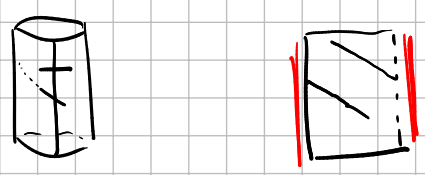
X der. at p if $X : \mathcal{C}^\infty(P, \mathbb{R}) \rightarrow \mathbb{R}$ linear and $X(fg) = f(p) X(g) + g(p) X(f)$

$$d\phi_p(A^*_p) f = A^*_p(f \circ \phi) = \frac{d}{dt}\left(f\left(\phi(p \triangleleft e^{tA})\right)\right)\Big|_{t=0}$$

$\phi(p \triangleleft e^{tA}) = \phi(p) \triangleleft e^{tA} = (u, g) \triangleleft e^{tA} = (u, g e^{tA})$

$$= \frac{d}{dt}\left(f(\phi(p) \triangleleft e^{tA})\right)\Big|_{t=0} = \frac{d}{dt}\left(f(u, \underbrace{g e^{tA}})\right)\Big|_{t=0}$$

$\rightarrow$ vertical because only changes in the group variable



choice of horizontal must play nice with group action

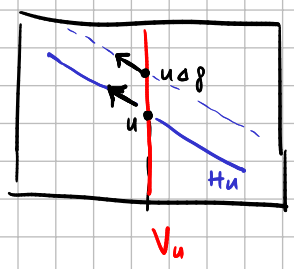
(group action on vector fields)

<u>requirement</u>: if X is horizontal vector field $(X_u \in H_u \ \forall u \in P)$

then $X \triangleleft g$ is still horizontal

(if $X_u \in H_u$ then $(dR_g)_u(X_u) \in H_{u \triangleleft g}$

<u>ex</u> $P = \mathbb{R}^2$ $(M = \mathbb{R}$ $G = \mathbb{R}$ (translation)$)$



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dual vector or covector is an elemt of $T_u^* P$

$\alpha \in T_u^* P$ means $\alpha : T_u P \to \mathbb{R}$ linear

1-form : covector field $\omega : P \to T^* P$ $\omega_u \in T_u^* P$

$u \mapsto (u, \omega_u)$

$\mathfrak{g}$- valued 1-form ?

$\mathrm{Lin}(V, W) \cong V^* \otimes W$

$\mathfrak{g}$-valued covector $\alpha : T_u P \to \mathfrak{g}$ linear <u>OR</u> $\alpha \in T_u^* P \otimes \mathfrak{g}$

$\mathfrak{g}$-valued 1-form map $\omega : u \in P \mapsto (u, \omega_u)$ ω_u $\mathfrak{g}$-valued covector

---

$TP = V \oplus H$

a given connection induces a $\mathfrak{g}$-valued 1-form ω defined like this

let $X_u \in T_u P$. then $\mathrm{proj}_{V_u}(X_u) = A_u^*$ for some $A \in \mathfrak{g}$ | $V_u = \{ A_u^* \mid A \in \mathfrak{g} \}$

$\boxed{\omega_u(X_u) = A \text{ by def.}}$

$A^*_u = 0 \iff A = 0$

Property if $X_u \in H_u$ then $\omega_u(X_u) = 0$

$\boxed{\text{Prop 1.1}}$ ⓐ $\omega(A^*) = A \quad \forall A \in \mathfrak{g}$

ⓑ $R_g^* \omega = g^{-1} \omega g \quad \forall g \in G$ (shorthand notation)

$\hookrightarrow (R_g^* \omega)(X) = g^{-1} \omega(X) g$

proof $v \in T_u P$

$$\overbrace{(R_g^* \omega)_u}^{\text{covector}}(v) \overset{\text{def. of pull back}}{=} \omega_{R_g(u)}\big((dR_g)_u(v)\big)$$

$$= \omega_{R_g(u)}\big((dR_g)_u(h) + (dR_g)_u(A^*_u)\big)$$

$$= \underbrace{\omega_{R_g(u)}\big(\underbrace{(dR_g)_u(h)}_{\in H_u}\big)}_{=0} + \omega_{R_g(u)}\big((dR_g)_u(A^*_u)\big)$$

$v = \underset{\in H_u}{h} + \underset{\in V_u}{A^*_u}$

$$\rightarrow (R_g^* \omega)_u(v) = \omega_{R_g(u)}\big((dR_g)_u(A^*_u)\big)$$

prop 5.1 $\quad A^* \triangleleft g = (g^{-1} A g)^*$

$$= \omega_{R_g(u)}\big((g^{-1} A g)^*_{R_g(u)}\big)$$

$$= g^{-1} A g$$

$g \in G \quad A \in \mathfrak{g}$

$\omega_u(A^*_u) = A$

$u \to R_g(u)$

$A \to g^{-1} A g$

$$= g^{-1} \omega_u(A^*_u) g$$

$$= g^{-1} \omega_u(h + A^*_u) g$$

$$= g^{-1} \omega_u(v) g$$

$\rightarrow R_g^* \omega = g^{-1} \omega g$

Converse if we have a $\mathfrak{g}$-valued 1-form ω on P that satisfies

- $\omega(A^*) = A$
- $R_g^* \omega = g^{-1} \omega g$

Then ω induces a connection || We call ω a <u>connection 1-form</u>

by def: if $s \in T_u P$ we say s horizontal if $\omega_u(s) = 0$

$$H_u = \ker(\omega_u)$$

ex: prove that $R_g^* \omega = g^{-1} \omega g \Rightarrow (dR_g)_u (H_u) \subseteq H_u$

$\pi : P \to M$ bundle proj

$d\pi_u : T_u P \to T_{\pi(u)} M$

$d\pi_u |_{H_u}$ is invertible $\Rightarrow H_u \cong T_{\pi(u)} M$